

Australia's Health Technology Assessment

How does the absence of an official cost-effectiveness (CE) threshold in Australia's Pharmaceutical Benefits Advisory Committee (PBAC) impact decision-making within the health technology assessment (HTA)

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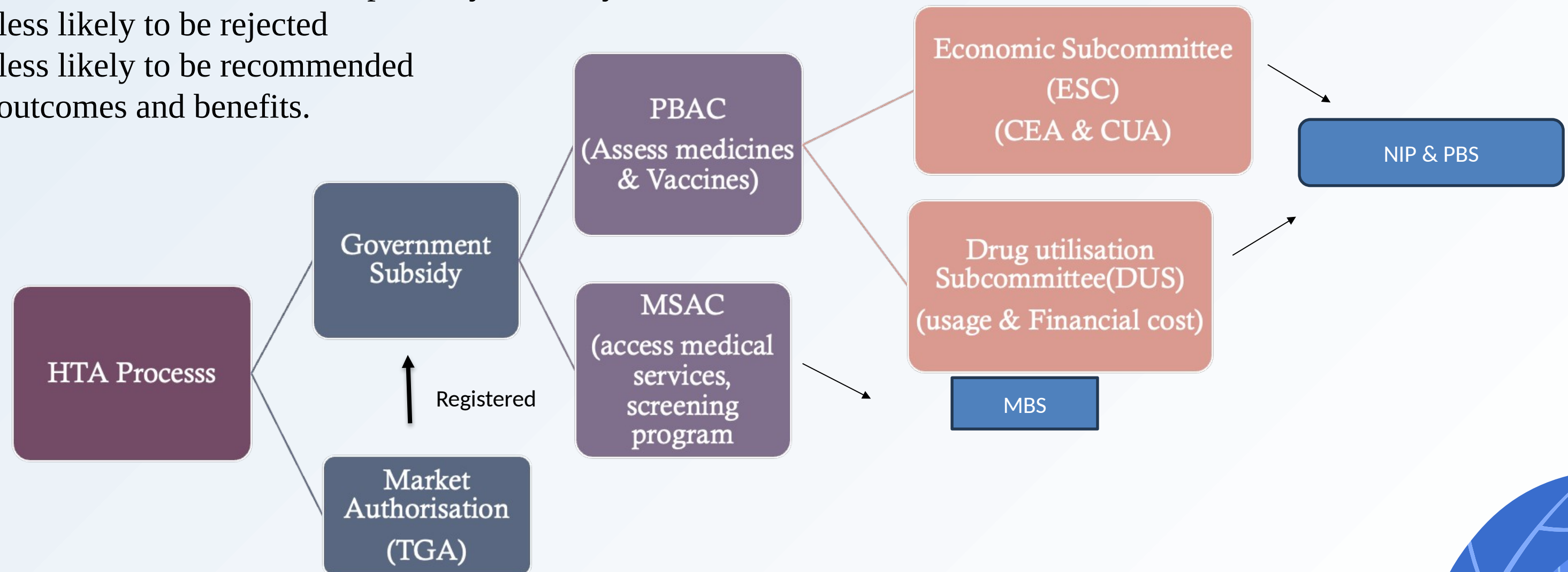


Australia Health Technology Assessment System Framework

- Evaluate quality, safety, and efficiency to provide affordable and equitable access to health intervention.
- Recommendation based on clinical and economic evaluation.
- Two HTA processes: Market Authorization and Government Subsidy (PBAC and MSAC)

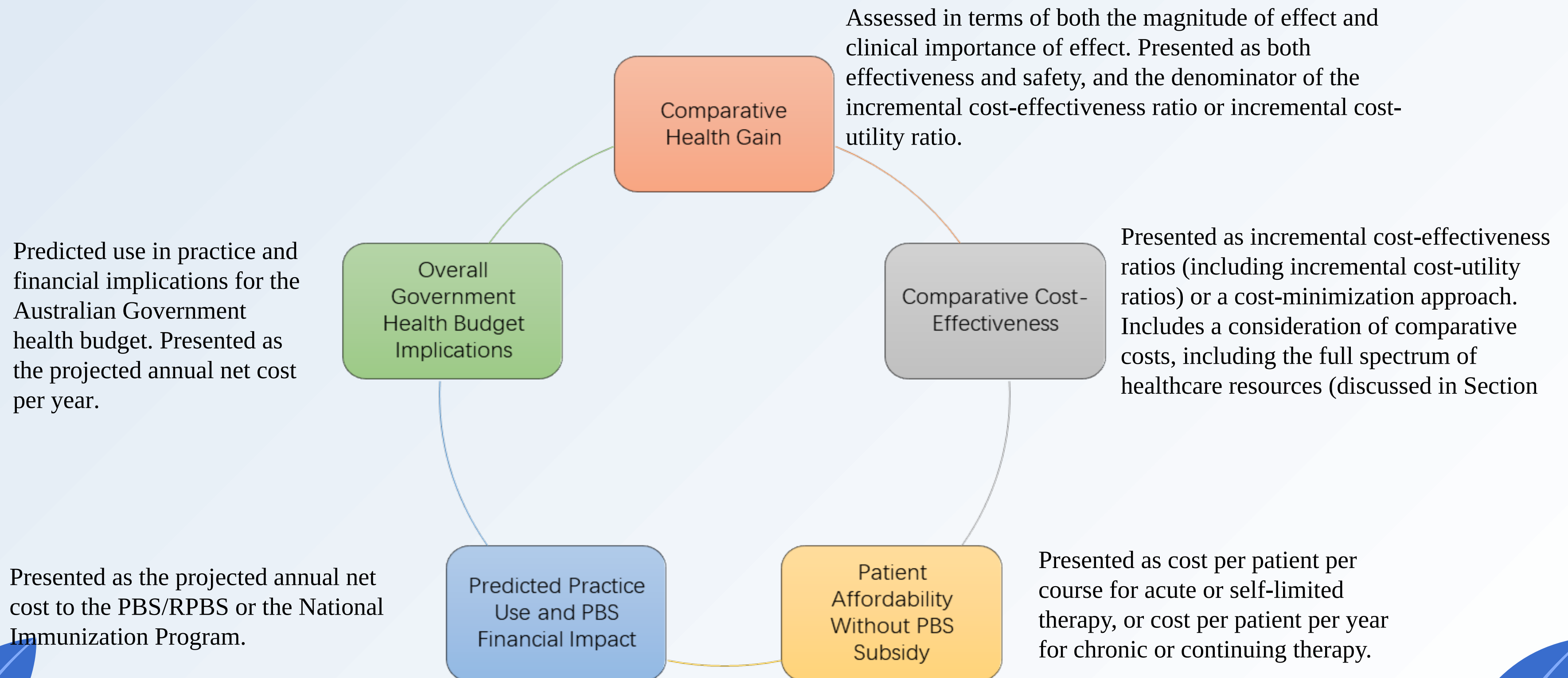
PBAC for PBS and NIP :

- Separate statutory body; that includes ESC and DUS
- Compare clinical effectiveness by RCT
- CUA is preferred over CEA, if a drug is clinically superior to the comparator.
- Implicit CEA threshold of ICER \$15000-\$45,000 per Qaly and Daly.
 - ICER < 45,000 are less likely to be rejected
 - ICER > 75,000 are less likely to be recommended
- Broader health system outcomes and benefits.



Five Quantitative Factors Influencing PBAC Decision-Making

According to Australian PBAC Guidelines, PBAC decision making is influenced by five quantitative factors:



- Drug name: 9-valent Human Papillomavirus (HPV) vaccine
- Time: 2017

Economic Analysis

(1) PBAC presented a cost-utility analysis comparing:

- A 9vHPV vaccine 2-dose program to the current 4vHPV vaccine 3-dose program.
 - A 9vHPV vaccine 3-dose program to the current 4vHPV vaccine 3-dose program.
- The 9vHPV vaccine schedule would result in more QALYs than the current 4vHPV vaccine schedule.

(2) ICER

the base case ICER for the 3-dose 9vHPV versus 3-dose 4vHPV vaccine schedules comparison of \$15,000/QALY - \$45,000/QALY was acceptable.



Component	9vHPV	4vHPV	Increment
9vHPV vaccine 2-dose NIP versus the current 4vHPV vaccine 3-dose NIP			
QALYs	16.550134	16.549669	0.000465
Incremental cost/QALY gained			Dominates
9vHPV vaccine 3-dose NIP versus the current 4vHPV vaccine 3-dose NIP			
QALYs	16.5501398	16.549669	0.000529
Incremental cost/QALY gained			Dominates

Source: July 2017 PBAC meeting

PBAC Outcome: The recommendation was made based on cost-effectiveness compared with a 3-dose schedule of 4vHPV vaccine.

Explicit Thresholds in UK vs. Implicit Thresholds in Australia

Brief intro of 9-valent vaccine in the UK

- The United Kingdom (UK) switched from using the 4-valent human papillomavirus (HPV) vaccine to the 9-valent vaccine in 2021.
- **Cost-effective analysis in UK:** An incremental cost-effectiveness ratio of **£8,600 per QALY** gained when discounted at 3.5% and **£3,300 per QALY** gained when discounted at 1.5%.
- **Explicit threshold :** In the UK, the National Institute for Health and Clinical Excellence (NICE) has a cost-per-QALY threshold of £20,000-£30,000.
- The results showed that the 9vHPV vaccine prevented a substantial additional number of HPV-related cases and deaths compared to the 4vHPV vaccine, and was **highly cost-effective**, with an incremental cost-effectiveness ratio of £8,600 per QALY gained when discounted at 3.5% and £3,300 per QALY gained when discounted at 1.5%.

Aspect	United Kingdom (UK)	Australia
Threshold Type	Explicit: £20,000-£30,000 per QALY gained	Implicit: No fixed threshold; ICER ranges are assessed for acceptability
Definition of Cost-Effectiveness	Determined explicitly by NICE guidelines	Evaluated flexibly based on PBAC's judgment of value for money and broader health system impacts
Metric Used	QALY (Quality-Adjusted Life Year)	QALY (Quality-Adjusted Life Year)
Decision-Making Approach	Relies heavily on NICE-defined explicit thresholds	Relies on case-specific evaluations; considers ranges like \$15,000-\$45,000 per QALY acceptable
Focus	- Health outcomes and quality of life improvement - Budgetary impact on the NHS	- Health outcomes, broader societal benefits - Tailored to Australia's healthcare priorities
Advantages	- Ensures transparency and consistency - Simplifies decision-making	- Considers comparative health gain and broader cost-effectiveness factors - Flexible to economic variations and government health budget constraints
Limitations	- Narrow focus on QALY as the primary metric - Limited due to rigid thresholds	- Lack of explicit thresholds may reduce comparability and standardization

Advantages of Implicit Thresholds in Australia's Health Technology Assessment

1. Flexibility

This approach allows PBAC to adjust its evaluations based on specific circumstances, without being restricted by fixed explicit thresholds.

- (1) the latest data
- (2) with special needs "rescue principle"

2. Multidimensional Evaluation

PBAC considers multiple factors, including clinical efficacy, safety, cost-effectiveness, and public health needs, rather than relying solely on a single numerical threshold.

3. Promotion of Innovation

The flexibility of the evaluation process provides greater acceptance for new drugs, ultimately driving public health progress.

A Support of Advantages

- Drug name: SELPERCATINIB
- Drug type and use: Non-small cell lung cancer (NSCLC)

PBAC recommended Selpercatinib as 2 key factors:

1. Comparative health gain

selpercatinib demonstrated superiority over pembrolizumab plus platinum-based doublet chemotherapy (a standard first-line treatment combining immunotherapy with two platinum-based chemotherapy drugs) in terms of progression-free survival (PFS), addressing both the magnitude and clinical importance of the health gain.

2. Comparative cost-effectiveness,

the PBAC considered selpercatinib would be cost-effective with the incremental cost-effectiveness ratio (ICER) of less than \$75,000 per QALY gained (using the effective price of pembrolizumab), consistent with other targeted therapies for NSCLC.

Disadvantages of Implicit Thresholds in Australia's Health Technology Assessment

1. Lack of Transparency

The absence of clear thresholds may make it difficult for external stakeholders to fully understand the evaluation results, leading to doubts about the fairness and consistency of PBAC's decisions, which can undermine public trust.

2. Uncertainty

1. Evaluation Bias:

There is a potential for evaluation bias, especially when data is insufficient or long-term effects have not been fully validated.

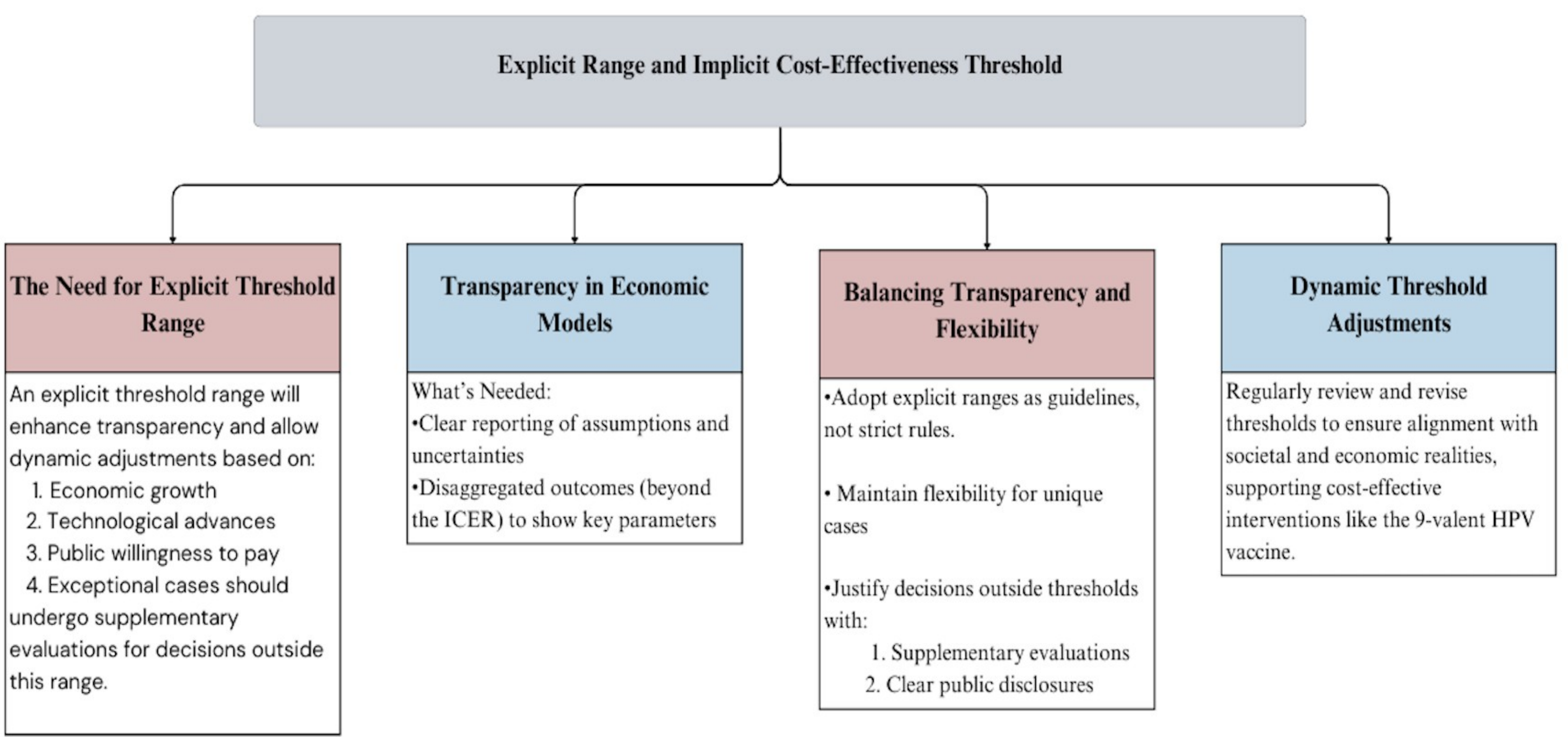
2. Complexity and Inefficiency:

Without fixed benchmarks, determining the cost-effectiveness and value of interventions becomes more complex and may result in inefficiencies.

3. Inconsistency

Different stakeholders may have divergent opinions due to the subjectivity of the standards, introducing the risk of bias and inconsistency in decision-making.

- George et al. (2001), found that even in the absence of an explicit CE threshold, the PBAC is broadly consistent with the use of economic efficiency as a criterion for deciding drugs that were recommended for listing.
- 'Herceptin Programme'
 - PBAC's positive recommendation of Herceptin for early-stage breast cancer, a possible application of the 'rule of rescue', resulted from intensive lobbying and media influence.



References

George, B., Harris, A. and Mitchell, A., 2001. Cost-effectiveness analysis and the consistency of decision making: evidence from pharmaceutical reimbursement in Australia (1991 to 1996). *Pharmacoeconomics*, 19, pp.1103-1109.

THANK YOU

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